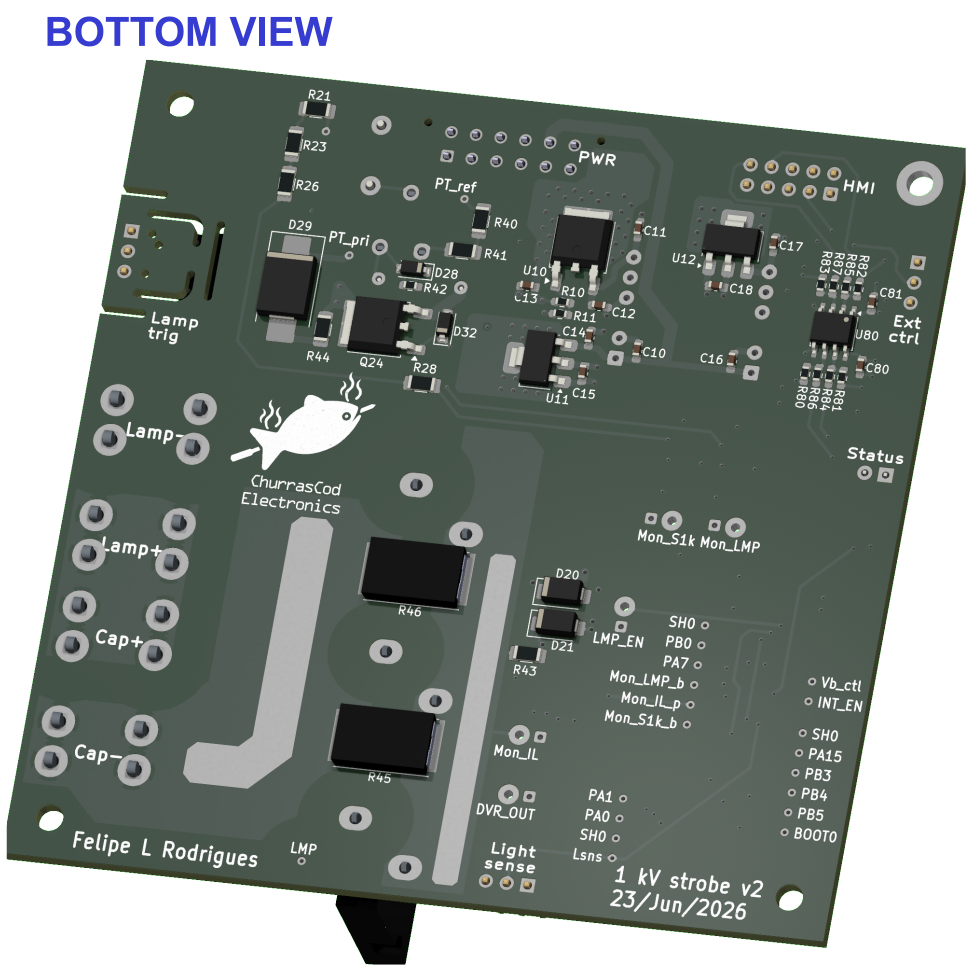
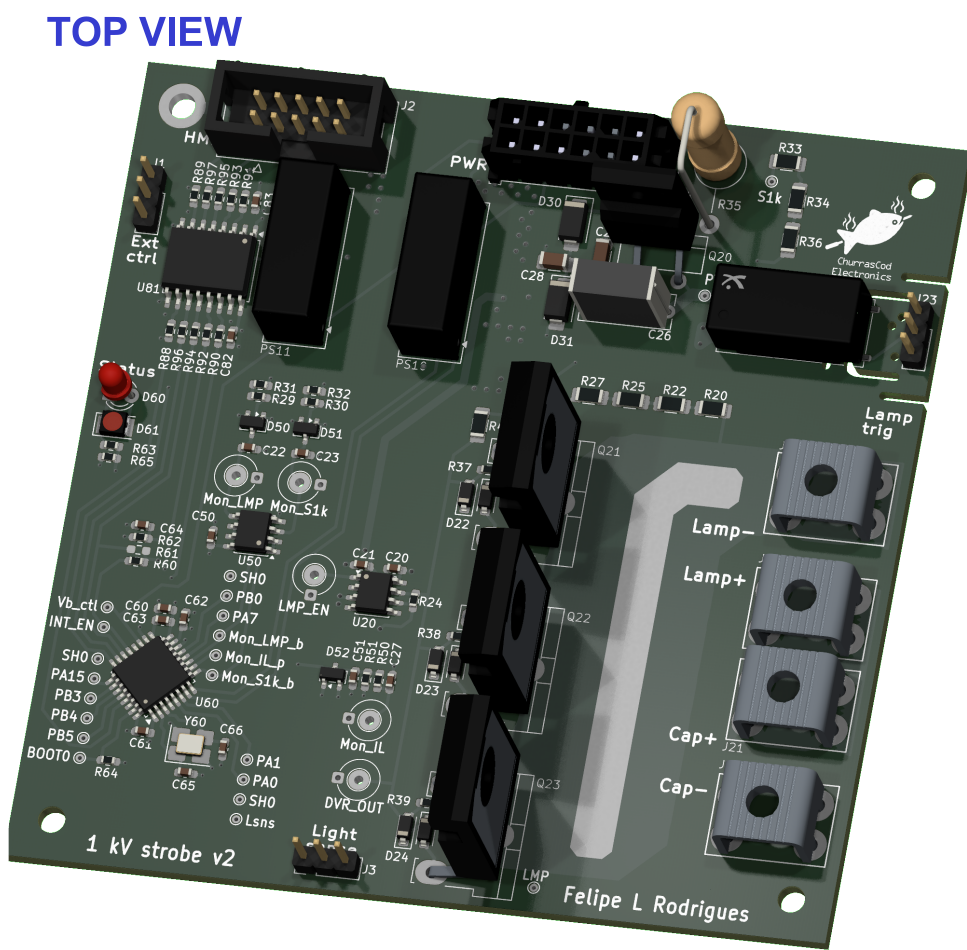


Main board

Rev 2.0.1

2026-09-05

Page	Index
1	Cover Page
2	Block Diagram
3	Power supply
4	Lamp driver
5	Signal conditioning
6	MCU
7	Optical isolation
8	Revision History

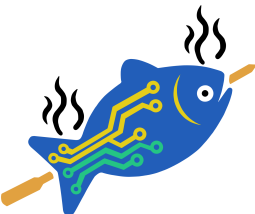


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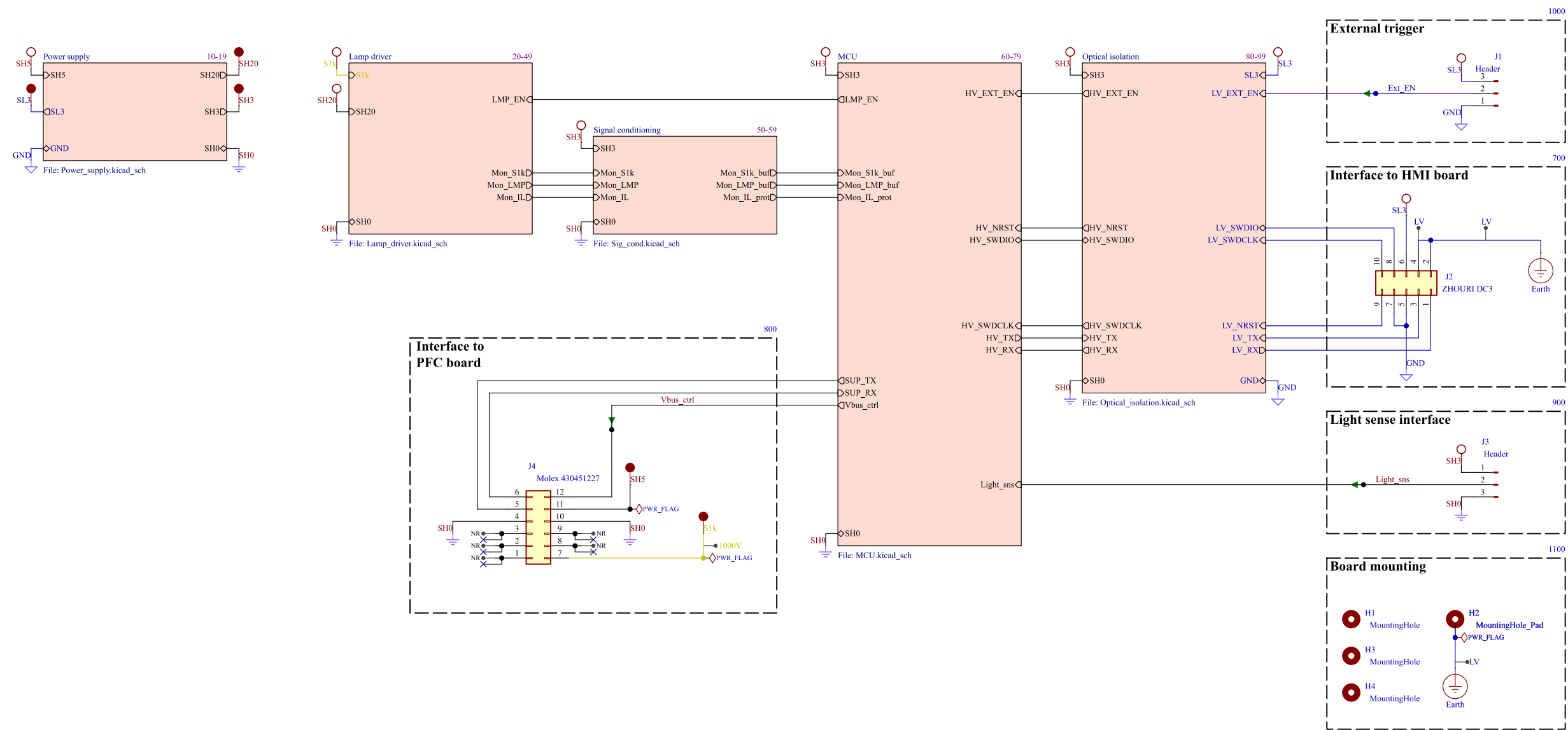
Date: 05-Sep-2026

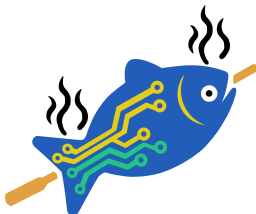
NOTE STYLING CONVENTION

DESIGN NOTE: Example text for informational design notes.	DESIGN NOTE: Example text for debug notes.	DESIGN NOTE: Example text for cautionary design notes.	DESIGN NOTE: Example text for critical design notes.	LAYOUT NOTE: Example text for critical layout guidelines.
--	---	---	---	--

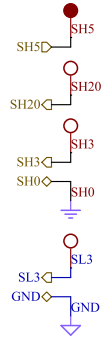
 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: RELEASED	Git Hash: 9c343a4
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title:		File Name: Strobe_main.kicad_sch	Designer: FR	Date: 2025-07-04	Revision: 2.0.1
	Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 8	

Block Diagram

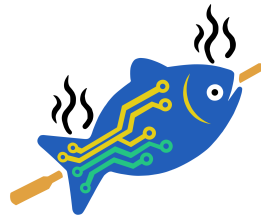
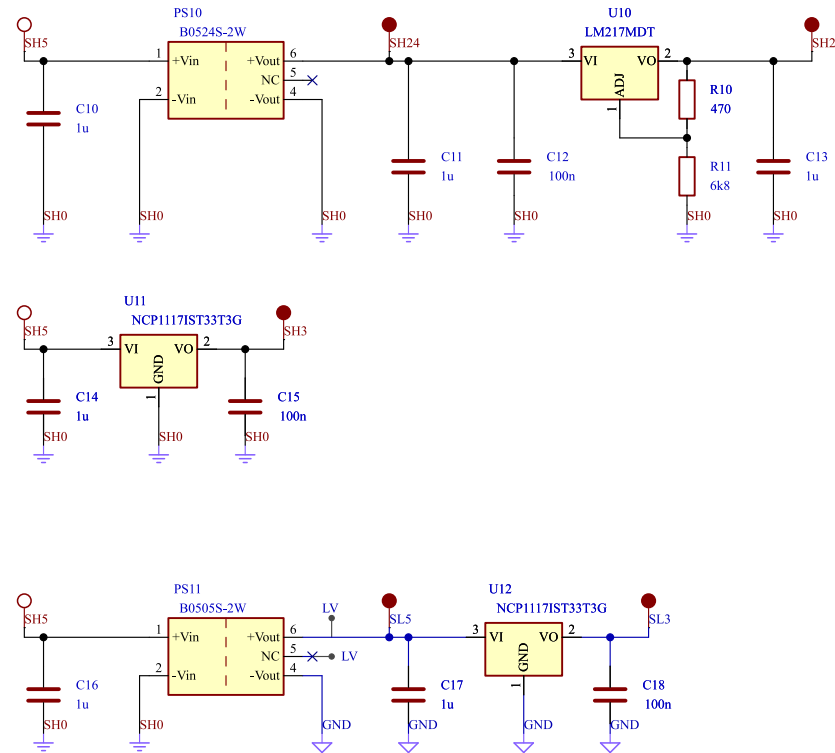


 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: RELEASED		Git Hash: 9c343a4			
			Board Name: Main board				Project Name: 1 kV strobe			
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-07-04		Revision: 2.0.1	
	Sheet Path: /Block Diagram/				Reviewer:		Size: A3		Sheet: 2 of 8	

BLOCK INTERFACE



Power supply 10-19



ChurrasCod
Electronics

Comments:

Sheet Title:

Power supply

Sheet Path:

/Block Diagram/Power supply/

Company:

Board Name:

Main board

File Name:

Power_supply.kicad_sch

Designer:

FR

Reviewer:

Variant:

RELEASED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

Git Hash:

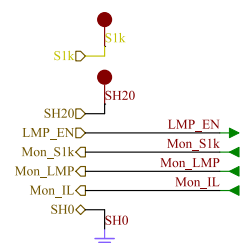
9e343a4

Revision:

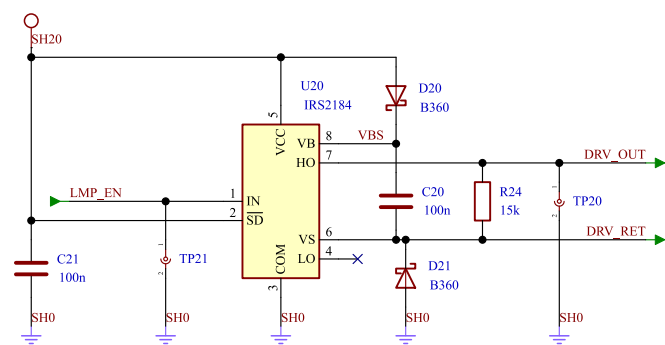
3 of 8

Lamp driver

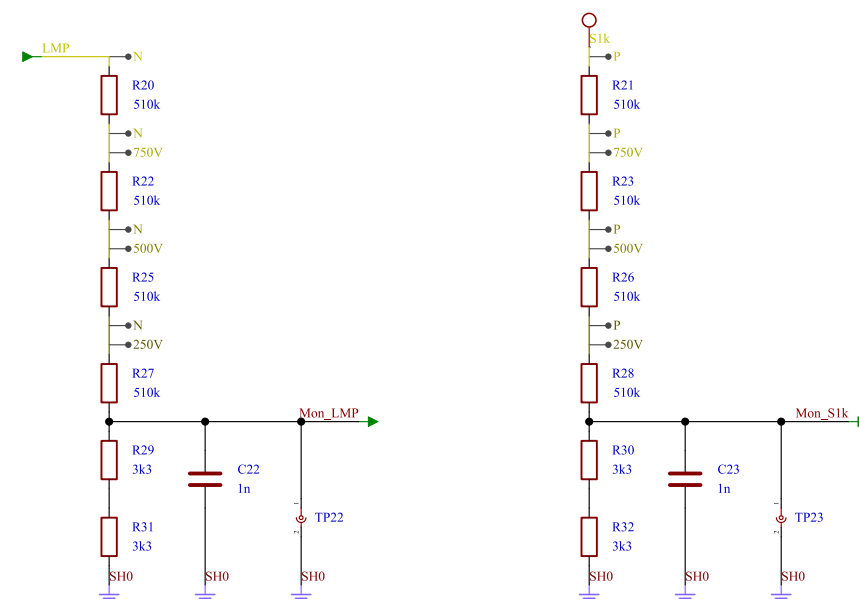
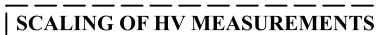
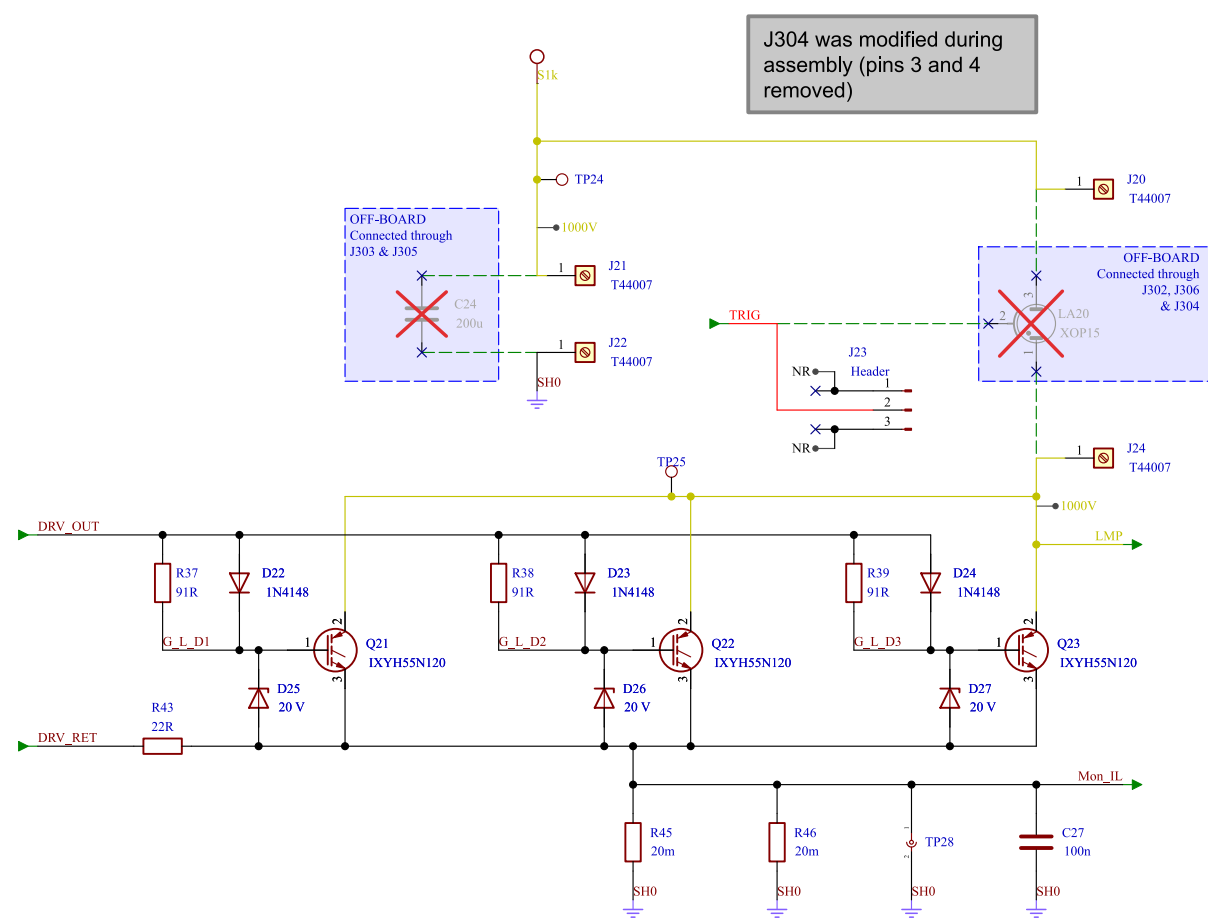
20-49



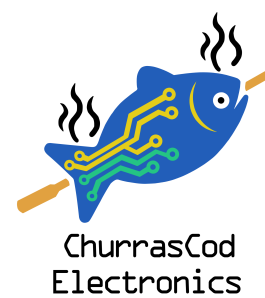
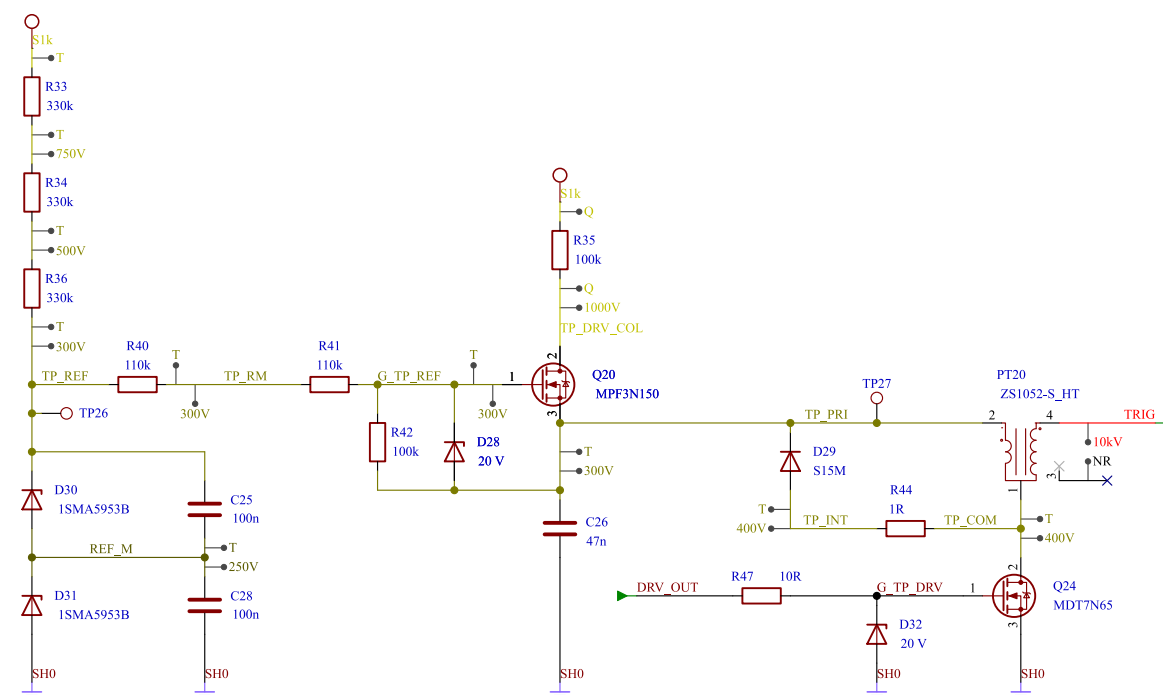
INPUT STAGE (IGBT DRIVER)



— — — — —
| **OUTPUT STAGE**

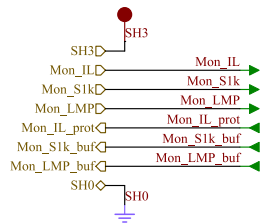


PULSE TRANSFORMER DRIVER

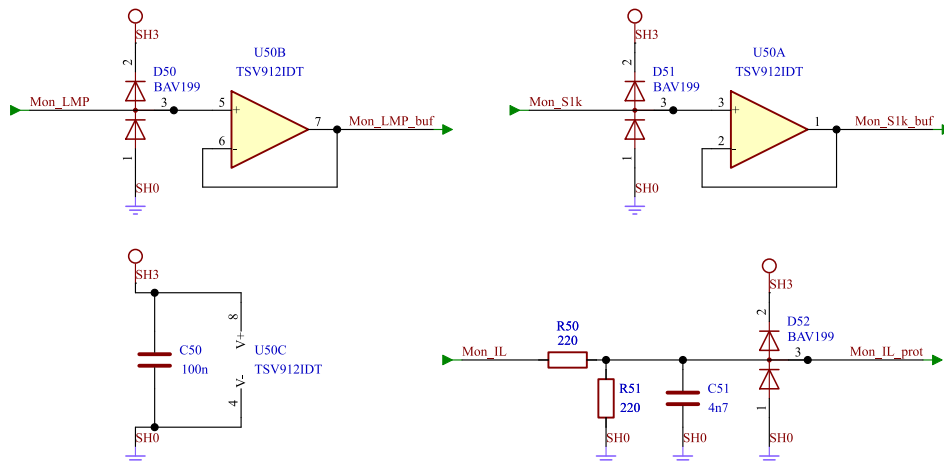


Comments:	Company:		Variant: RELEASED	Git Hash: 9c343a4
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Lamp driver	File Name: Lamp_driver.kicad_sch	Designer: FR	Date: 2025-07-04	Revision:
Sheet Path: /Block Diagram/Lamp driver/		Reviewer:	Size: A3	Sheet: 4 of 8

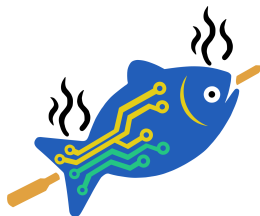
BLOCK INTERFACE



LAMP DRIVER VOLTAGES & CURRENT



Signal conditioning 50-59



ChurrasCod
Electronics

Comments:

Sheet Title:

Signal conditioning

Sheet Path:

/Block Diagram/Signal conditioning/

Company:

Board Name:

Main board

File Name:

Sig_cond.kicad_sch

Designer:

FR

Reviewer:

Variant:

RELEASED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

Git Hash:

9e343a4

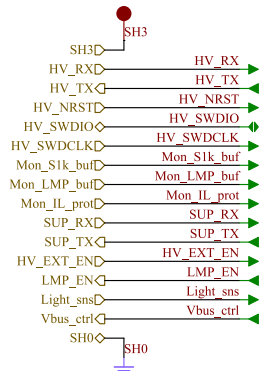
Revision:

5

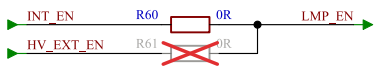
Sheet:

of 8

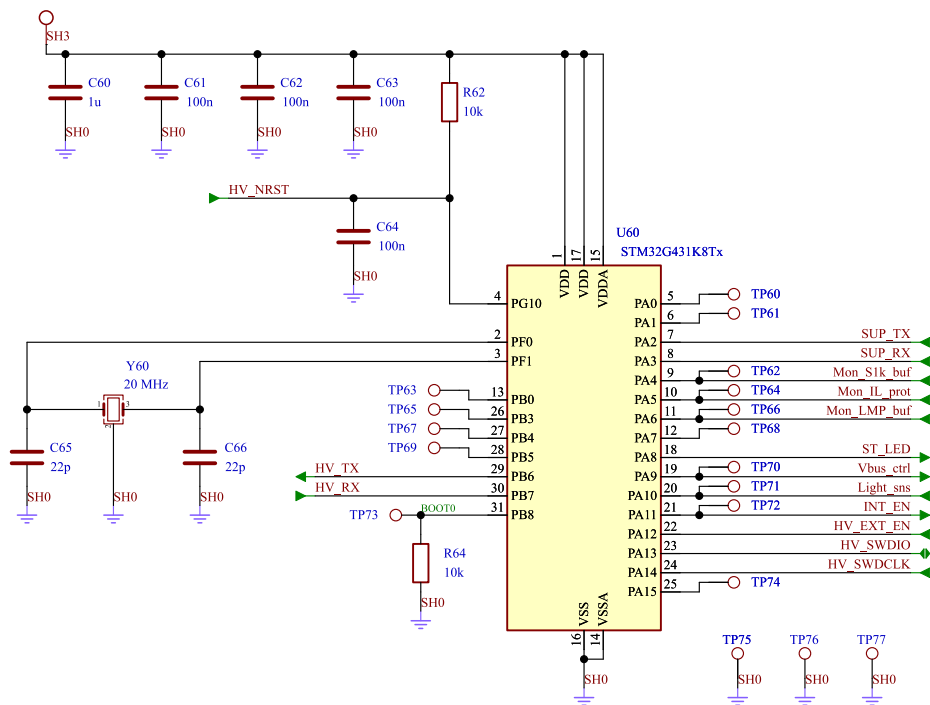
BLOCK INTERFACE



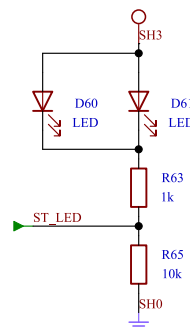
Trigger source selection



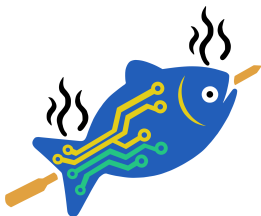
Main controller



Status LED



MCU
60-79



ChurrasCod
Electronics

Comments:

Sheet Title:

MCU

Sheet Path:

/Block Diagram/MCU/

Company:

Board Name:

Main board

File Name:

MCU.kicad_sch

Designer:

FR

Reviewer:

Variant:

RELEASED

Project Name:

1 kV strobe

Date:

2025-07-04

Size:

A4

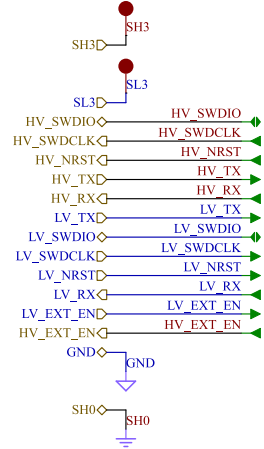
Git Hash:

9e343a4

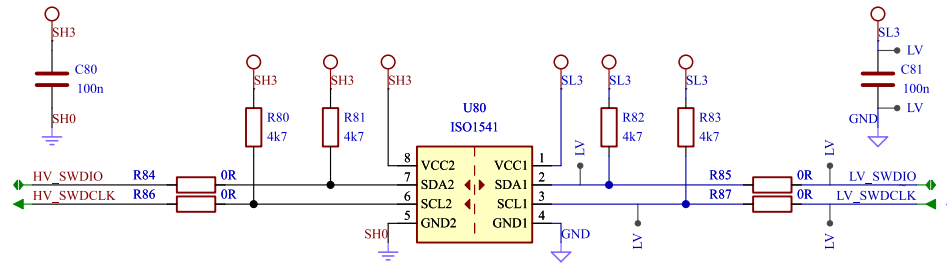
Revision:

6 of 8

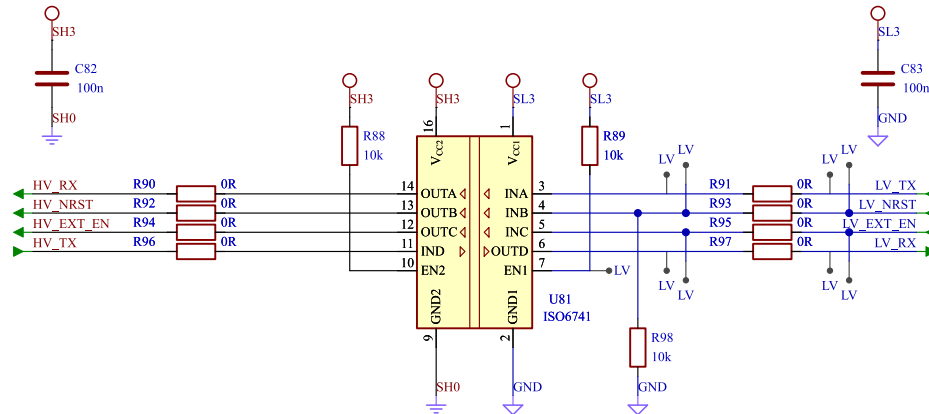
BLOCK INTERFACE



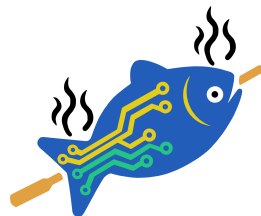
SWD optical isolator



NRST(SWD) + UART optical isolator



Optical isolation 80-99



ChurrasCod
Electronics

Comments:

Company:

Variant:

Git Hash:

RELEASED

9e343a4

Board Name:

Main board

Project Name:

1 kV strobe

Sheet Title:

Optical isolation

File Name:

Optical_isolation.kicad_sch

Designer:

FR

Date:

2025-07-04

Revision:

Sheet Path:

/Block Diagram/Optical isolation/

Reviewer:

Size:

A4

Sheet:

7 of 8

Revision History

Version 2.0.1 - 2026-09-05

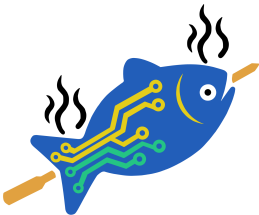
- Changed C22 and C23 to 1 nF

Version 2.0.0 - 2026-06-23

- decreased impedance of Ipeak measurement divider, removed buffer and fed signal directly to MCU
- reworked gate drive resistors
- reworked power input
- modified some of the testpoints and added extra "GND testpoints"
- reworked test pulse circuit to achive higher frequency
- added mlcc to MCU rst line

Version 1.0.0 - 2026-04-11

Migrated existing project to template (KDT)



ChurrasCod
Electronics

Comments:	Company: ChurrasCod Electronics		Variant: RELEASED	Git Hash: 9e343a4
	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-07-04	Revision: 2.0.1
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8